System Capacity and Care Load Analytics for Unaccompanied Children: A Time Series Monitoring and Forecasting Framework

Kunal Kashyap
Machine Learning Engineering Intern
Email: kunalkash50@gmail.com

***Abstract*—The Unaccompanied Alien Children (UAC) Program operates as a dynamic care pipeline involving intake, transfer, sheltering, and discharge across multiple federal agencies. Effective system operation requires continuous monitoring of care capacity and inflow-outflow balance. However, operational reporting often lacks structured analytics for identifying sustained stress periods and forecasting future load.**

This study proposes a modular analytics framework for monitoring and forecasting healthcare capacity within the UAC system. The framework integrates data validation, feature engineering, rolling trend analysis, capacity stress detection. An interactive Streamlit dashboard was developed to support real-time monitoring and decision support.

Results : show that rolling averages and net intake indicators effectively identify sustained pressure periods, while forecasting models provide early signals for potential system overload. The proposed framework transforms raw operational data into actionable insights for proactive humanitarian response planning.

***Index Terms*—Healthcare Capacity Analytics, Operational Monitoring, Stress Detection, Streamlit Dashboard**

I. Introduction

Humanitarian and healthcare systems require proactive monitoring to prevent operational overload and delayed service delivery. The Unaccompanied Alien Children (UAC) program functions as a multi-stage pipeline involving intake by U.S. Customs and Border Protection (CBP), transfer to Department of Health and Human Services (HHS) facilities, medical and psychological support, and eventual discharge to vetted sponsors.

Managing such systems requires real-time awareness of total care load, balance between inflow and outflow, and sustained capacity pressure. Traditional reporting methods provide descriptive summaries but lack structured analytics capable of detecting emerging stress patterns.

This research develops an analytical framework that transforms operational data into interpretable capacity indicators and forecasting insights.

II. Problem Statement

Although daily operational data is available, there is no centralized analytical framework capable of continuously assessing:

- Total system care load
- Balance between intake and discharge
- Capacity stress periods
- Future system load levels

Without such analytics, decision-making becomes reactive and may lead to overcrowding, delayed care, and operational strain.

III. Dataset Description

The dataset consists of daily operational records describing the flow of children through the care system.

Column	Description
Date	Reporting date
CBP Daily Intake	Children apprehended and placed in CBP custody
CBP Custody	Total children currently in CBP custody
Transfers to HHS	Children transferred from CBP to HHS
HHS Care	Children currently under HHS care
HHS Discharges	Children discharged to sponsors

TABLE I
Dataset Variables

IV. Data Validation

Before analytical modeling, the dataset undergoes validation procedures including:

- Detection of missing reporting dates
- Duplicate record identification
- Logical constraints validation

Two primary constraints are enforced:

$$Transfers \leq CBP\ Custody \tag{1}$$

$$Discharges \leq HHS\ Care \tag{2}$$

Records violating these constraints are flagged rather than removed to preserve transparency.

V. Feature Engineering

To translate raw operational counts into interpretable metrics, several derived indicators are computed.

A. Total System Load

$$Total\ System\ Load = CBP\ Custody + HHS\ Care \quad (3)$$

This metric represents the total number of children under system responsibility.

B. Net Daily Intake

$$Net\ Intake = Transfers - Discharges \quad (4)$$

Positive values indicate increasing pressure on the care system.

C. Care Load Growth Rate

$$Growth\ Rate = \frac{Load_t - Load_{t-1}}{Load_{t-1}} \quad (5)$$

This measures the rate at which system burden changes over time.

VI. Trend and Stress Analysis

Daily operational data often contains noise that obscures long-term patterns. To address this, rolling averages are applied.

A. Rolling Average

$$RollingMean = \frac{1}{n}\sum_{i=1}^{n} Load_{t-i} \quad (6)$$

7-day and 14-day windows are used to detect sustained pressure.

B. Stress Window Detection

Capacity stress windows are defined as periods where:

- Net intake remains positive over multiple days
- Rolling load exceeds median system load

These periods indicate prolonged operational strain.

VII. Dashboard Implementation

An interactive dashboard was developed using Streamlit and Plotly to support operational monitoring.

Key features include:

- Date range filtering
- KPI summary cards
- Rolling trend visualization
- Net intake monitoring
- Stress window highlighting

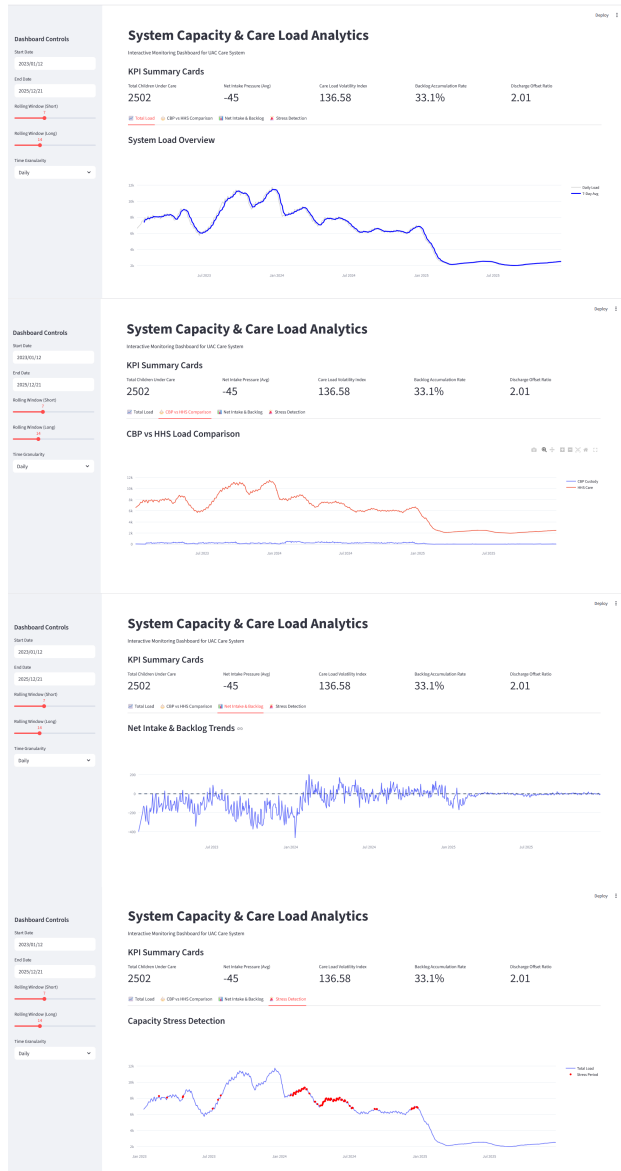

Fig. 1. Streamlit Capacity Monitoring Dashboard

VIII. Results

The analytical framework reveals several operational insights:

- Sustained positive net intake leads to backlog accumulation
- Rolling averages reveal pressure patterns not visible in daily data
- Volatility spikes correspond to unstable operational periods

Forecasting models provide early warning signals for potential capacity strain.

IX. Conclusion

This research presents a scalable framework for monitoring and forecasting capacity within humanitarian care systems.

By integrating operational analytics, time series modeling, and interactive dashboards, the framework enables proactive decision-making and improved situational awareness.

The combination of ARIMA and Prophet models provides both statistical robustness and interpretability.

X. Future Work

Future improvements may include:

- Seasonal ARIMA,SARIMA And Prophet modeling
- External event regressors
- Automated model selection
- Real-time alert systems

Acknowledgment

The author acknowledges the use of publicly available operational datasets and open-source analytics tools.

References

[1] S. Kagne, "Mentor and Guide for this Project," Available: https://saikagne.github.io/

[2] K. Kashyap, "System Capacity & Care Load Analytics for Unaccompanied Children – Project Repository," GitHub Repository. Available: https://github.com/kunal-kashyap635/UACproject